Software Engineering LAB 1 Submission

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Class : AIML 5-G

Requirements Table

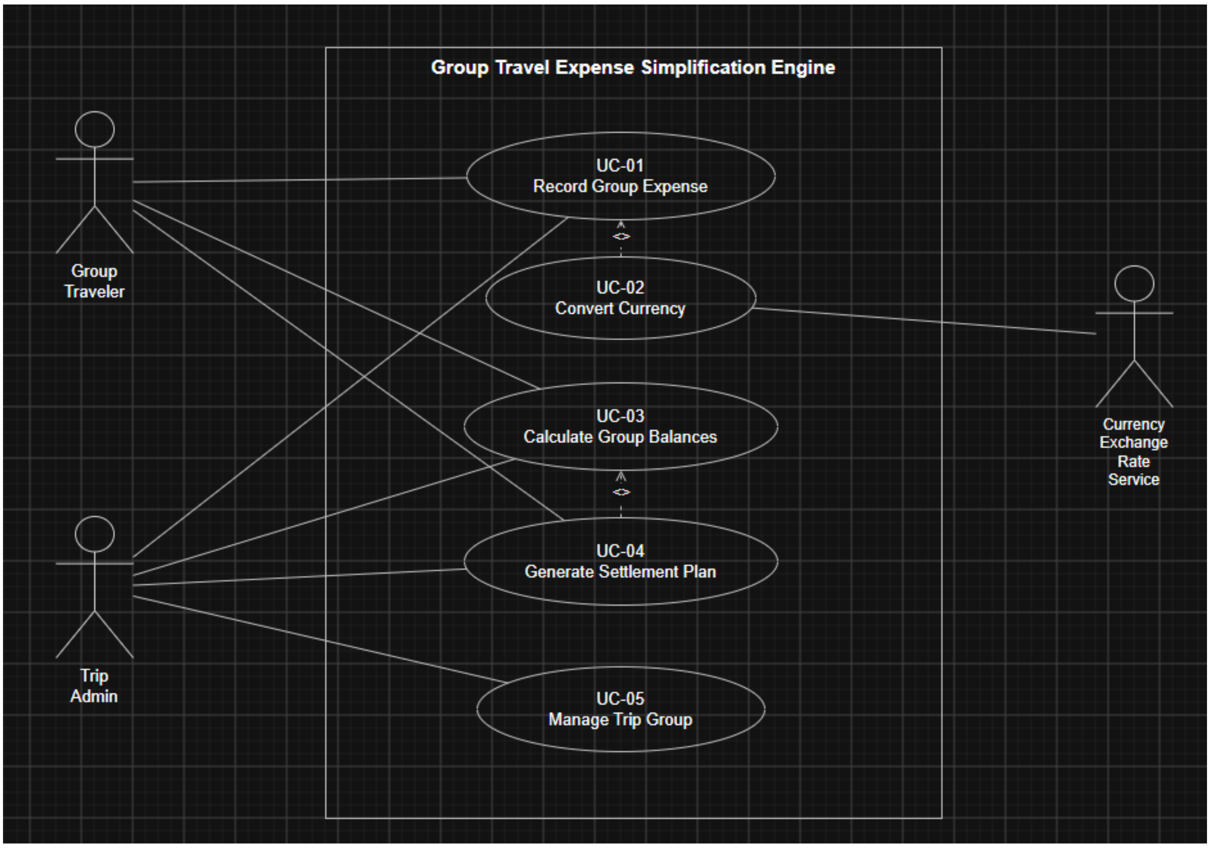
Problem Statement #37 — Group Travel Expense Simplification Engine

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute the optimal debt settlement plan using a minimum-flow algorithm, minimizing the total number of peer-to-peer payment transactions needed to settle a group's balances.	High	Pass: computed balances net to zero and the plan uses the minimum possible number of payment transactions. Fail: sum of debts does not equal sum of credits, or a plan with fewer transactions exists.	Fewer transactions make it faster and simpler for travelers to actually settle up after a trip.
FR-002	Functional	The system shall allow a group traveler or trip admin to record a group expense, including amount, currency, payer, and the travelers the expense is shared among.	High	Pass: a submitted expense with valid amount, currency, payer, and participants appears in the trip's expense list with all entered details. Fail: expense is rejected, or a required field is missing after submission.	All balance and settlement calculations depend on expenses being recorded correctly.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-003	Functional	The system shall allow a group traveler or trip admin to split a recorded expense unevenly across travelers instead of dividing it equally.	High	Pass: entered custom split amounts sum to the total expense amount and are saved per traveler. Fail: split amounts do not sum to the total, or the system only allows an equal split.	Group expenses are often shared unequally, so an equal-only split would produce incorrect balances.
FR-004	Functional	The system shall convert any expense recorded in a foreign currency to the trip's base currency, using the exchange rate for the date the expense was recorded.	High	Pass: converted amount equals expense amount multiplied by that date's exchange rate, within 0.01 units of the base currency. Fail: conversion uses the wrong date's rate, or differs by more than 0.01 units.	A common currency is needed before balances across a multi-currency trip can be compared or settled.
FR-005	Functional	The system shall calculate each traveler's net balance (amount owed or owed to them) from all recorded expenses and their converted amounts.	High	Pass: for each traveler, balance equals total amount paid minus total share of expenses, and all travelers' balances sum to zero. Fail: balances do not sum to zero, or a recorded expense is not reflected in any balance.	Net balances must be known before a settlement plan can be generated.

Req ID	**Type**	**Description**	**Priority**	**Acceptance Criteria**	**Rationale**
NFR-001	Nonfunctional	The settlement calculation engine shall execute in under 100 ms for a group of up to 50 travelers with 500 recorded expenses.	High	Pass: benchmark tests with 50 travelers and 500 expenses show settlement computation completing in under 100 ms. Fail: computation exceeds 100 ms in any benchmark run.	Travelers expect the settlement plan to appear almost instantly when requested, even for larger groups.
NFR-002	Nonfunctional	The system shall use exchange rate data that is no more than 24 hours old for every currency conversion.	Medium	Pass: every conversion log entry shows an exchange-rate timestamp within 24 hours of the conversion time. Fail: any conversion is performed using a rate older than 24 hours.	Stale exchange rates would make converted amounts, balances, and the settlement plan inaccurate.

UML Use-Case Diagram:



Use-Case Flow: Generate Settlement Plan

Group Travel Expense Simplification Engine — Problem Statement #37

Use Case ID	UC-04
Use Case Name	Generate Settlement Plan
Primary Actor	Group Traveler (Trip Admin may also trigger this use case)
Secondary Actor(s)	None
Preconditions	The trip group exists with at least one traveler. At least one group expense has been recorded (UC-01), and any foreign-currency expenses have already been converted to the trip's base currency (UC-02).
Postconditions	A settlement plan showing the minimum set of peer-to-peer payments needed to bring every traveler's balance to zero is generated and displayed to the user.

Main Success Scenario

1. The group traveler selects “Generate Settlement Plan” for the trip.

2. The system includes Calculate Group Balances (UC-03) to compute each traveler's net balance from all recorded and currency-converted expenses.

3. The system runs the min-flow settlement algorithm on the net balances to determine the smallest possible set of payments that settles the group.

4. The system verifies that the total debts in the settlement plan equal the total credits (net balance of zero).

5. The system displays the settlement plan, listing who pays whom and how much.

6. The use case ends successfully.

Alternate Flow — Incomplete Expense Data

3a. While calculating balances in step 2, the system finds an expense with a missing payer or an invalid amount.

3a1. The system stops the settlement calculation and shows an error listing the incomplete expense(s).

3a2. The group traveler or trip admin is asked to correct or remove the flagged expense(s).

3a3. The use case ends without producing a settlement plan.